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Epidermal medical applications

brought about a revolution in both the doctors' and patients' lives. These help us to save so many lives. Older patients often complain about being confined to the hospitals. Devices like these are a great rescue in such cases. Although the initial equipment costs are currently high in the market, I strongly believe that in a country like India, where not enough doctors are available to treat everywhere and all the time, adaptation of such technologies will lead to lower costs in a couple of years."

Dr Surabhi Arora, consultant ophthalmologist, explains, "Often concerns are raised about data security. The technology provider assures us that it is highly encrypted, but patients still find it hard to trust. Trust is one of the biggest hindrances in the adaptation of these technologies. In the western world, companies like IBM and Microsoft have long been providing health assistance to hospitals, mostly through AI and other customer support facilities. Most of the results have been positive so far. In India, we are still in the very initial stages and have a long way to go."

"Amazon has funded a cancer detection startup called Grail, while Apple has acquired a sleep tracking technology maker called Beddit. Most hospitals and patient care centres use Microsoft's cloud platform to store and analyse their data. Such success stories provide a big boost for medical professionals, as it reinforces positivity in adopting the new developed unconventional path for treating patients," shares Dr Surabhi.

Mentioning the use of augmented reality in vein visualisation, Paul Pickering says, "Venipuncture, the

technique of puncturing a vein to draw blood or deliver an intravenous injection, is one of the most common medical procedures. However, there are extra challenges in case of some patients such as the elderly, burn victims, drug abusers and patients undergoing chemotherapy. Of the three million procedures performed daily in the US, an estimated 30 per cent require multiple attempts before finding a suitable vein."

"Augmented reality can help. USA-based AccuVein uses non-invasive infrared (IR) technology to scan the target site and display the underlying vein structure. Because hemoglobin in the blood absorbs more red light than the surrounding tissues, the resulting image shows veins as a web of black lines on a background of red," he adds.

India's medicare: On the high-tech route

Jangir shares, "Wireless technologies mainly being used are Wi-Fi, GSM and Bluetooth. So, while designing the boards, we ensure to integrate modules for these technologies. For instance, users are adopting ECG devices with Bluetooth connectivity to access ECG data through their phones."

There are applications of Zigbee as well. Globally, use of Zigbee is quite widespread in medical equipment. In India, a handful of brands use Zigbee for medical equipment as their Wi-Fi is already occupied by other applications.

In addition to becoming wirelessly connected, many devices are turning portable. One good example is ultrasound devices. **EFY**